

A review of PKN-type modeling of hydraulic fractures

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ABSTRACT

Many numerical modeling techniques have been employed to understand the phenomena occurring during hydraulic fracturing. Despite the various complex models developed, such as the planar three-dimensional and pseudo three-dimensional models, the one-dimensional Perkins–Kern–Nordgren (PKN) model is still of great interest because it serves as a starting point for the development of more efficient and robust algorithms. This work reviews all PKN-type models from the first published model in 1961 to recent advancements in the model design. The underlying assumptions of the fundamental physical processes of the PKN model are first analyzed. Developments of and notable contributions to PKN models are then summarized into four main groups and discussed in a relatively chronological order. The differences between the standard modeling approach that employs the fluid flux as a dependent variable and the other that utilizes the particle velocity are highlighted. The trend of accounting for the poroelastic effect in the leak-off process and the effect of turbulence in the flow regime is also discussed. A number of challenges in modeling and computations caused by the strong nonlinear and degenerate governing system and how they have been tackled are then discussed in detail. Finally, some open suggestions for future work are mentioned.

1. Introduction

Hydraulic fracturing is a well operation widely used in the oil and gas industry to enhance the recovery of hydrocarbons in reservoirs. Other applications include weakening and fragmenting the rockmass in mining activities (Zhang et al., 2002, 2017) and drill–cutting disposal of waste (Moschovidis et al., 2000). These fractures also naturally occur as kilometer-long dikes, where magma breaks through the crust to the surface of the earth (Spence and Turcotte, 1985; Lister, 1990).

The Perkins–Kern–Nordgren (PKN) model, named after three contributing researchers (Perkin, Kerns, and Nordgren) is one of the most widely used hydraulic fracture models in the oil and gas industry. Since its introduction, more complex models, such as the pseudo three-dimensional (P3D) or the planar three-dimensional (PL3D) models, have been introduced for practical applications. However, one-dimensional (1-D) models, such as the PKN (Nordgren, 1972), Khristianovic–Geertsma–de Klerk (KGD) (Zheltov, 1955; Geertsma and De Klerk, 1969), and radial or penny-shaped model (Sneddon, 1946) are still widely used by researchers in the hydraulic fracturing field. This is

due to the complex nature of hydraulic fracturing processes; to understand the nature of propagation regimes and develop efficient computational algorithms, the hydraulic fractures of simple geometries are first investigated. Gaining an understanding of 1-D models and successfully developing efficient algorithms for them is a crucial step in solving more realistic models.

Among the three classical models (PKN, KGD, and penny-shaped models), the PKN (also called the blade-shaped or finger-like) model is of practical importance in the petroleum industry mainly because of its underlying assumption that the fracture length is much greater than the fracture height. Because the minimum in situ stress in most reservoirs is horizontally directed, a crack easily propagates perpendicular to the minimum stress, leading to vertical-oriented hydraulic fractures. Additionally, layers blocking the upward and downward growth of hydraulic fractures often bound reservoir layers. As a result, the propagation velocity in the horizontal direction is much higher than that in the vertical direction and numerous occurrences of blade-like geometries occur. This model is, thus, applicable in a wide range of reservoirs, as indicated in the broad field data by De Pater (2015). As a result, most P3D models used in the industry are extensions of the PKN model in which the

Abbreviations: BVP, Boundary value problem; GMS, Gaukler–Manning–Strickler; KGD, Khristianovic–Geertsma–de Klerk; HFTS, Hydraulic Fracture Test Site; PDE, Partial differential equation; PKN, Perkins–Kern–Nordgren; PL3D, Planar three-dimensional; P3D, Pseudo three-dimensional; 1-D, One-dimensional.

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Nomenclature			
E	Young's modulus, Pa	$t_0(x)$	Time at which the fluid passes through the point x , sec
E'	Plane strain modulus, Pa	s	Distance from the tip, m
$E(\cdot)$	Complete elliptic integral of the second kind	w	Fracture width at the center of the fracture, m
G	Shear modulus, Pa	$\bar{w}$	Averaged fracture width
K_{IC}	Fracture toughness, $Pa.m^{\frac{1}{2}}$	w_*	Length scale for fracture width
M	Number of time steps	w^D	Fracture width reduction
M_c	Fracture compliance, $m.Pa^{-1}$	x	Horizontal coordinate, m
N	Number of nodal points	y	Vertical coordinate, m
Q_0	Constant injection rate at the inlet, m^3/sec	δL	Relative error of the crack length
Re	Reynolds number	$\delta \hat{w}$	Relative error of the crack opening
$\bar{Re}$	Equivalent Reynolds number	$\delta \hat{\varphi}$	Relative error of the reduced particle velocity
c	Diffusivity coefficient, m^2/sec	ε	A small distance behind the fluid front
$f(t^*)$	Influence function	η	Poroelastic coefficient
f_B	Blasius friction factors	μ	Fluid viscosity, $Pa.sec$
f_D	Darcy friction factor	ν	Particle velocity
f_M	GMS friction factors	$\bar{\nu}$	Average fluid velocity
f_L	Laminar friction factor	ξ	Dimensionless/self-similar horizontal coordinate
g	Fluid leak-off velocity, $m.sec^{-1}$	ξ_0	Point where the integral in equation (5) becomes a singularity
h	Fracture height, m	ξ_ε	Dimensionless horizontal position at ε
k	Wall roughness	ξ_*	Dimensionless/self-similar fluid front position
l	Fracture length, m	Π	Dimensionless net pressure
n	Flow behavior index	ρ	Fluid density, $kg.m^{-3}$
p	Net pressure, Pa	σ_0	Minimum principal stress normal to the plane of the fracture, Pa
p_f	Fluid pressure in the fracture, Pa	ν	Poisson's ratio
p_0	Pore pressure, Pa	ψ	Self-similar width variable
q	Fluid flux, m^2/sec	φ	Reduced particle velocity
t^*	Dimensionless fracture surface exposure time	Ω	Dimensionless fracture width

fracture height varies in time and space (Settari and Cleary, 1986; Adachi et al., 2010; Zhang et al., 2017; Cohen et al., 2017). Furthermore, analytical and semi-analytical solutions of PKN-type models have been used as benchmarks and test grounds for more complex P3D- and PL3D-type models (Ribeiro and Sharma, 2013; Zhang et al., 2017; Zia and Lecampion, 2019).

This work reviews and summarizes the developments of and notable contributions to PKN-type models since their introduction in 1961 into four chronological and developmental groups. The first group includes the classical model that brings the name to the PKN model, ranging from the pioneering work by Perkins and Kern (1961) until the study by Kemp (1990). The second group includes models using a fluid flux approach motivated by the comprehensive research on the behavior of the near-tip region and propagation regimes of hydraulic fractures by a group of researchers from the Schlumberger Cambridge Research and the University of Minnesota (Desroches et al., 1993, 1994; Lenoach, 1995; Garagash and Detournay, 2000). The third group is composed of models using the particle velocity approach to solve the PKN model, including the work by Linkov (2011b) that was developed further by him and his colleagues. The fourth and final group of models, which takes into account the turbulent effect of fluid flow, has been getting more attention due to recent usage of high-flow, low-viscosity slickwater in hydraulic fracturing treatments (Ames and Bunger, 2015; Dontsov, 2016; Zia and Lecampion, 2017).

For a thorough understanding of various critical issues of the physics of hydraulic fracturing, readers are suggested to refer to other review articles. In particular, the singular behavior at the near-crack tip region caused by strong coupling of fluid lubrication and elastic solid deformation was discussed by Detournay (1999). He also analyzed the propagation regimes of hydraulic fractures and parameters controlling these regimes in impermeable rocks (Detournay, 2004). Adachi et al. (2007) and Lecampion et al. (2017) discussed issues and challenges in

the existing mathematical modeling and computational simulation of hydraulic fractures. Explicit and implicit numerical methods in hydraulic fracturing using the particle velocity as a primary quantity were reviewed in detail by Linkov (2019). Readers can also refer to (Economides and Nolte, 2000) for a general understanding of the various aspects of a hydraulic fracturing operation in the petroleum industry. Lastly, readers should follow the series work in the grand project of hydraulic fracture test site (HFTS) (Ciezobka et al., 2018) to understand the practical behavior of hydraulic fracturing process by conducting a comprehensive core analysis including petrophysics, geomechanics, microseismic methods. Analytical results showed a complex network of interaction between hydraulic fractures and natural fractures compared with modeling simplicity; it also showed how stress variation among the rock layers and rock heterogeneity affect the distribution of proppant placement (Maity et al., 2018; Maity and Ciezobka, 2019).

Before reviewing PKN models, the underlying assumptions behind the classical PKN model (Perkins and Kern, 1961; Nordgren, 1972; Kemp, 1990) are presented and analyzed. The four groups of PKN models are then discussed in a relatively chronological order. Computational challenges of the PKN problem and how they have been addressed are then discussed. Finally, research gaps are discussed in the form of future work.

2. Assumptions behind the classical PKN model

The 1-D classical PKN problem is assumed to propagate along the x -axis and is completely contained in a horizontal permeable layer with an elliptical vertical cross-section and a constant height h , as depicted in Fig. 1. Here, $w(x, t)$ is the fracture width at the center of the fracture ($y = 0$), $l(t)$ is fracture length, Q_0 is the constant injection rate at the inlet, μ is the dynamic viscosity of the fracturing fluid, and $q(x, t)$ is the average flow rate per unit height of the fracture (i.e., fluid flux). Additionally,

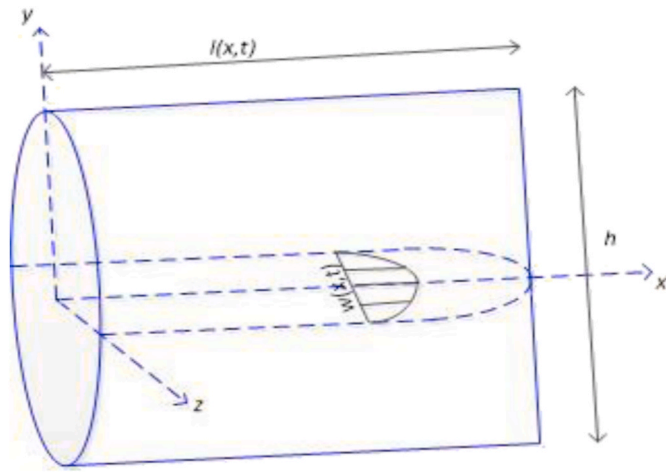


Fig. 1. Schematic of the PKN model of a hydraulic fracture. h , $l(x, t)$, $w(x, t)$ are the fracture height, length, and width, respectively.

$p(x, t) = p_f(x, t) - \sigma_0$ is the net pressure, where p_f represents the fluid pressure in the fracture and σ_0 represents the minimum principal stress normal to the plane of the fracture. The elastic properties of the rock are characterized by Young's modulus E and Poisson's ratio ν .

Hydraulic fracturing is a complex coupled mechanism fundamentally describing the rock deformation induced by the internal fluid pressure, the fluid flow inside the fracture, and the leaking of fluid in the fracture into the surrounding host rock. In the classical PKN model, the rock deformation follows the theory of elasticity, the fluid flow in the fracture follows Poiseuille's law, and the leak-off process follows Carter's leak-off rate (Carter, 1957), as summarized in Fig. 2. The incorporation of these processes into the continuity equation results in a governing strongly nonlinear Reynold's lubrication equation of the problem. In this section, the implications behind these theoretical processes are discussed.

2.1. Elastic deformation of the rock

Three types of loading are generally considered to enable crack propagation: i) opening (tensile loading), ii) sliding (in-plane shear loading), and iii) tearing (out-of-plane shear loading). The PKN model assumes negligible shear stresses and that the fracture, thus, propagates only owing to the tensile stress, as has been demonstrated in hydraulic fracturing field operations (Maxwell and Cipolla, 2011).

The PKN elasticity equation was derived from Sneddon's solution (1946) for a dry thin crack (i.e., Griffith crack) to describe the local relation between a fracture's net pressure and its opening, expressed as

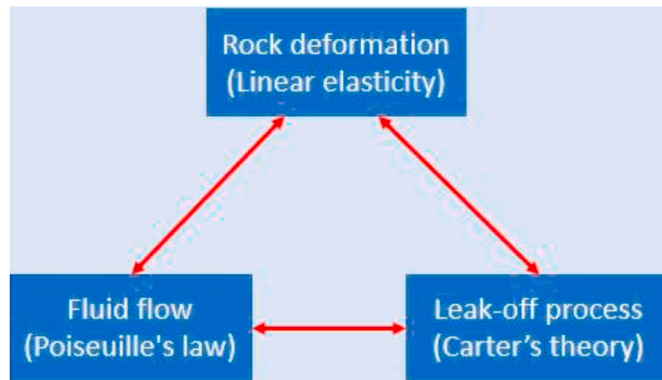


Fig. 2. Interaction between the fundamental physical processes of the PKN model.

$$w = \frac{2h}{E} p \quad (1)$$

where $E' = \frac{E}{1-\nu^2}$ representing the plane strain modulus.

Employing this elasticity equation means that the width of the fracture opening at a given point depends only on the net pressure at that point. This feature of PKN geometry is often referred to as vertical plane strain elasticity and is a consequence of assuming a small height-to-length ratio. Flaws related to employing the local elasticity equation are discussed in Section 3.

The nature of propagation regimes of hydraulic fractures has been well documented (Detournay, 2004). In the case of no leak-off, two energy dissipative mechanisms govern the hydraulic fracture: i) viscous dissipation and ii) energy release owing to the fracture propagation. If the viscous dissipative energy is dominant, then the fracture propagates in the viscous-dominated regime. Otherwise, it propagates in the toughness-dominated regime. The classical PKN model does not consider the fracture toughness, K_{IC} , thus assuming that the fracture propagates in the viscous-dominated regime in the horizontal direction.

2.2. Fluid flow in the fracture

The flow of fluid in a PKN-type fracture is assumed to follow the lubrication theory (Poiseuille's law). The Poiseuille flow equation is derived from the application of the Navier–Stokes equations to a flow between two parallel plates. Under laminar flow conditions, liquid in a narrow channel between two plates experiences viscosity resistance. To balance this resistance, the fluid pressure in the narrow channel (i.e., the hydraulic fracture) decreases as the distance from the injection point increases. The governing fluid flow equation relates the fluid pressure gradient to the fracture width expressed in the form of a nonlinear partial differential equation (PDE).

In particular, this assumption means that fluid flow in the PKN model is one-dimensional flow directed along the horizontal axis in an elliptical channel. The fluid pressure p_f is assumed to be constant at each vertical cross-section orthogonal to the propagating direction (i.e., $p_f = p_f(x, t)$). The PKN model assumes an incompressible Newtonian fluid. The formulation for fluid flow in the PKN model is, thus, expressed as

$$q = -\frac{\pi h w^3}{64\mu} \frac{\partial p}{\partial x} \quad (2)$$

As this equation is highly nonlinear, it poses computational and numerical challenges when solving the PKN problem; these issues are discussed in Section 4.

2.3. Leak-off process

As fluid leaks from a hydraulic fracture into the porous host formation, a thin and low-permeability layer (i.e., the filter cake) forms on the fracture walls. It is the result of the deposition of polymer molecules from the leak-off fluid. An invaded zone due to the filtrate is then created in the surrounding porous medium. In the PKN model, fluid loss from the hydraulic fracture is assumed to be one-dimensional in a direction normal to the axis of the propagating crack. This diffusion process is considered time-dependent and pressure-independent and is usually referred to as Carter's leak-off (Carter, 1957). The formula is written as

$$g(x, t) = \frac{C'}{\sqrt{t - t_0(x)}} \quad (3)$$

where g represents the fluid leak-off velocity estimating the two fracture walls, t is the lapsed time since pumping began, $t_0(x)$ is the time at which the fluid passes through the point x , and $C' = 2C_L$, with C_L denoting Carter's leak-off coefficient.

3. Model summary

In this section, PKN models developed from 1961 to date are briefly reviewed to highlight the evolution of the PKN problem modeling without covering all of the findings in detail. The considered studies include complete numerical models, analytical asymptotic solutions, and particular aspects crucial to the PKN problem. These models are divided into four groups based on the approaches with which they solve the governing equations and in a relatively chronological order. Group 1 is the classical PKN model. Group 2 includes models that solve the governing equations using the fluid flux and the fracture width as primary variables. Some models that consider the effect of poroelasticity are also discussed. Group 3 includes models that are solved using the fluid particle velocity as a primary quantity. Finally, group 4 includes models that consider the effect of turbulent fluid flow in the fracture.

3.1. Group 1: classical PKN models

The PKN model was first introduced in the pioneering work by Perkins and Kern (1961), based on Sneddon's solution of a dry line crack (Sneddon, 1946) in a vertical hydraulic fracture with a fixed height and an elliptical cross-section, propagating in the longitudinal axis of the fracture only. The formula used to calculate the fracture width was proposed by coupling a plane strain elasticity equation and a viscous fluid flow equation (based on the lubrication theory of viscous fluid flows between two narrow parallel plates). In this original model, fluid leak-off from the fracture to the host rock and volume change (storage) owing to the changing of the fracture width were not considered, and no formula for crack length was provided. Later, Nordgren (1972) made a significant contribution to Perkins and Kern's work by considering 1-D fluid loss that behaves according to Carter's law (Carter, 1957) (i.e., the leak-off velocity is proportional to the inverse square root of the time elapsed since the beginning of the leak-off process) and fracture volume change. The two effects were added in the form of a continuity equation. Additionally, numerical solutions for the nonlinear partial differential governing equations and analytical asymptotic solutions for the two cases of leak-off (large time) and no leak-off (small time) were provided. Fracture mechanics and the crack tip effect were assumed to be negligible.

In his study of the (non-dimensional) governing equations of Nordgren's model, Kemp (1990) noted a missing Stefan condition essential to the nature of this moving boundary problem. This Stefan condition states that the average velocity of the fluid at the fluid front is equal to the speed at which the fluid front itself moves. In the case of no leak-off, the self-similar solution indicates that crack tip width behaves as the one-third power of the distance $(1 - \xi)^{\frac{1}{3}}$ from the crack tip (where $\xi = \frac{x}{l(t)}$). By comparing with Nordgren's numerical finite difference solution, he proved Nordgren's model failed to produce this proper behavior $((1 - \xi)^{\frac{1}{3}})$. Thus, the Stefan condition is necessary to describe accurately the tip of the crack in the numerical solution of the PKN model. Furthermore, he extended the model to shut-in cases for both no leak-off and high leak-off. Kemp also used the fourth degree of the crack width (w^4) rather than the crack width (w) itself as a variable in the numerical computations. Like Nordgren's model, Kemp's model only considered the fracture propagation because of the viscous dissipation of the fluid flow and did not address the effect of fracture mechanics.

Before proceeding with other groups, the complete set of governing equations for the classical PKN model is presented here, which helps explain the differences and advancement in the following models.

Fracture width equation (follows linear elasticity):

$$w = \frac{2h}{E} p \quad (4)$$

Flow equation for Newtonian fluid (follows Poiseuille's law):

$$q = -\frac{\pi h w^3}{64\mu} \frac{\partial p}{\partial x} \quad (5)$$

Leak-off equation (follows Carter's law)

$$g(x, t) = \frac{C^*}{\sqrt{t - t_0(x)}} \quad (6)$$

Local mass balance (also called continuity equation):

$$\frac{\partial q}{\partial x} + \frac{\partial w}{\partial t} + g = 0 \quad (7)$$

Total mass balance (obtained by integrating the local mass balance, both in time and space):

$$\frac{Q_0 t}{2h} = \int_0^{l(t)} w dx + 2C^* \int_0^{l(t)} \sqrt{t - t_0(x)} dx \quad (8)$$

3.2. Group 2: fluid flux approach

This group is called the fluid flux approach because the governing system in these models is derived and solved using the fluid flux as a dependent variable (unlike group 3, in which the governing system is derived in terms of the particle velocity). This group is subdivided into models that consider the effect of poroelasticity induced by leak-off or the leak-off velocity is not dependent on Carter's theory, and models that consider the asymptotic behaviors of the crack tip region and propagation regimes.

A PKN hydraulic fracture model accounting for the poroelasticity based on an explicit method was first developed by Detournay et al. (1990). The model was formulated in a moving coordinate system and made use of a fixed mesh and an explicit finite difference scheme in numerical solving. The poroelastic effect on the fracture width is quantified by a reduction amount w^p , expressed as

$$w^p = -2\eta \Delta p M_c f(t^*) \quad (9)$$

where η is a poroelastic coefficient, $0 \leq \eta \leq 1/2$; M_c is fracture compliance, defined as $M_c = \pi(1 - \nu)h/4G$ where G is shear modulus; $\Delta p = p_f - p_0$ where p_0 is the pore pressure; t^* is a dimensionless fracture surface exposure time, defined as $t^* = \frac{4c(t - \tau(x))}{h^2}$, c is the diffusivity coefficient; and $f(t^*)$ is an influence function (not given explicitly for general case) varying between 0 and 1 as t^* changes from 0 to ∞ , respectively. Equivalently, the net pressure reduction owing to poroelasticity is $2\eta \Delta p f(t^*)$.

This study showed that the poroelastic effect caused a notable decrease in the net pressure; however it had a slight change in fracture width and almost no change in fracture length. The unchanged fracture length can be explained by the assumed pressure-independent leak-off coefficient. The difference between the injection and leak-off volumes, which is unaffected by the poroelasticity effect, governs the fracture length. Abousleiman (1991) further developed this model by accounting for the pressure-dependent leak-off equation instead of Carter's leak-off theory using the same finite difference method in a moving coordinate. The work was extended for the prediction of the pressure decline in the wellbore after the fracture closure. For a pressure-dependent leak-off, the prediction of both fracture dimensions and pressure were considerably affected by poroelasticity. Although these models considered the effect of poroelasticity, they did not consider the effect of the crack tip region to the fracture propagation and the solution accuracy of the PKN problem.

This, thus, led to the development of models that properly accounted for the asymptotic behaviors of the tip region and fluid propagation regimes to obtain better accuracy in numerical solutions. For example, Adachi and Peirce (2008) proposed a novel formula to remedy a shortcoming of using the local elasticity equation in the classical PKN model. As shown before by Kemp (1990), the plane strain (local) elasticity

assumption leads to the fracture width near the tip region behaving as $(1 - \xi)^{\frac{1}{3}}$, which is incompatible with the classical square-root asymptote of fluid pressure. This feature constrains the classical PKN model to situations when the rock toughness is negligible. Thus, they derived a non-local elasticity equation for the PKN model from the general elasticity equation of a planar fracture (Hills et al., 1996), retaining the assumptions that the vertical cross-section is elliptical and fluid pressure is independent from the y coordinate. The non-local elasticity equation expresses the relationship between the dimensionless net pressure Π and dimensionless fracture width Ω in the fracture as

$$\Pi(\xi) = -1/\pi \int_{-1}^1 \Omega'(\xi_0) G(\xi, \xi_0; \beta) d\xi_0 \quad (10)$$

here, the integral is singular in the Cauchy principal value sense and take its Cauchy principal value at $\xi = \xi_0$, and the kernel G is defined as

$$G(\xi) = \text{sgn}\left(\xi_0 - \xi\right) \sqrt{1 + \frac{\beta^2}{(\xi_0 - \xi)^2}} E\left(\frac{\beta^2}{(\xi_0 - \xi)^2 + \beta^2}\right) \quad (11)$$

$$\beta = \frac{H}{L}; \Pi = p \frac{H}{E w^*}; \Omega = \frac{w}{w^*}; \xi = \frac{x}{l} \quad (12)$$

where $E(\cdot)$ represents the complete elliptic integral of the second kind, $\text{sgn}(\cdot)$ denotes the sign of the argument, $(\cdot)'$ indicates the derivative with respect to the argument, and w^* denotes the proper length scale for the fracture width.

Kovalyshen and Detournay (2009) re-examined the PKN model by scaling the arguments and performing a perturbation analysis of crack tip asymptotic behaviors. From scaling schemes, there exists two similarity solutions of the PKN hydraulic fracture: a storage-viscosity-dominated solution (referred to as the M solution) for a small time and leak-off-viscosity-dominated solution (referred to as the $\bar{M}$ solution) for a large time. The PKN tip behavior was characterized by the storage-dominated asymptotic solution in the near field (i.e., the tip aperture w behaves according to $w \propto s^{\frac{1}{3}}$, where s is the distance from the tip), and by the leak-off-dominated asymptotic solution in the far field (i.e., w behaves according to $w \propto s^{\frac{2}{3}}$). Thus, there is always a boundary layer at the crack tip where the leak-off is negligible and $w \propto s^{\frac{1}{3}}$, regardless of the overall propagation regime of the fracture. This is a result of the assumption of plane strain elasticity and might be remedied using a non-local elasticity equation, as suggested by Adachi and Peirce (2008).

Adachi et al. (2010) studied the PKN model as a special case of the classical P3D when the fracture height is contained within the reservoir. The model was implicitly solved using a collocation scheme with a fourth-order accuracy, which approximated the average crack opening and fluid flux using a cubic polynomial over each discretized interval. The permissible large size of each time step sped up the simulation over many temporal orders of magnitude so that it could be employed with a rather coarse mesh (i.e., a mesh with the number of nodal points $N = 10$). Furthermore, a different scaling approach that was used in previously published works (Detournay, 2004; Kovalyshen and Detournay, 2009) was adopted. The time, space, and field variables were expressed in terms of a constant quantity and a dimensionless variable; this led to a more rigorous scaled system and could be used to explicitly determine the condition of the fracture height growth (i.e., the fracture height is contained in the reservoir, partially/completely penetrates in the bounding layers, and unstable).

Dontsov and Peirce (2016) later investigated employing three toughness criteria within the PKN model. In the first two, the effect of lateral toughness was captured by a pressure boundary condition estimated based on either energetic considerations (i.e., $p(l) = \frac{2}{\sqrt{\pi H}} K_{IC}$), which was proposed by Nolte (1991), or on a radial fracture tip solution (i.e., $p(l) = \sqrt{\frac{\pi}{2H}} K_{IC}$), which was proposed by Sarvaramini and Garagash

(2015). The third replaced the plane strain elasticity equation with a non-local elasticity equation first proposed by Adachi and Peirce (2008). The results indicated that the employ of the non-local elasticity provided accurate approximations of fracture width, length, and net pressure for a large range of fracture toughness values (i.e., $0.16 \leq K_{IC} \leq 1.57 \text{ MPa}\cdot\text{m}^{\frac{1}{2}}$ for a reference set of parameters $h = 0.05\text{m}$, $\mu = 30.2\text{Pa}\cdot\text{s}$, $\nu = 0.4$, $E = 3.3\text{GPa}$, $Q_0 = 1.7\text{mm}^3/\text{s}$, $\Delta\sigma = 4.3\text{MPa}$, $t = 604\text{s}$). Additionally, the first two criteria using local elasticity were deemed applicable for moderate values of fracture toughness (i.e., $K_{IC} < 0.47\text{MPa}\cdot\text{m}^{\frac{1}{2}}$ for the same reference set of parameters).

3.3. Group 3: particle velocity approach

This approach, introduced by Linkov in a series of published works (2011a, 2011b, 2011c, 2012a, 2012b), employs the particle velocity (rather than the fluid flux) and fracture width considered as a degree as dependent variables for the modified governing equations. The particle velocity v is the velocity of fluid particles through the fracture cross-section (i.e., $v = q(x, t)/w(x, t)$). Using the particle velocity facilitates the use of the speed equation (also called the Stefan condition) to trace the fracture front. The Stefan condition for hydrofracturing problem was first presented by Spence and Sharp (1985) for a plane strain problem and later for PKN geometry by Kemp (1990) but then was abandoned for few decades until Linkov reintroduced this concept.

Linkov revisited the Nordgren problem in the case of no leak-off for studying a new approach and introduced the following self-similar variables:

$$x = \xi t^{\frac{4}{3}}; w(x) = t^{\frac{1}{3}} \psi\left(x t^{-\frac{4}{3}}\right); x_* = \xi_* t^{\frac{4}{3}} \quad (13)$$

where the asterisk (*) denotes the position of the fluid front, ξ is the self-similar horizontal coordinate, ψ is the self-similar width variable and t is time.

The governing system was, thus, reduced to an ordinary differential equation (ODE):

$$\frac{d^2 y}{d\xi^2} + a\left(y, \frac{dy}{d\xi}, \xi\right) - \frac{3}{20} = 0 \quad (14)$$

$$\text{where } y(\xi) = \psi^3(\xi) \text{ and } a\left(y, \frac{dy}{d\xi}, \xi\right) = \frac{\frac{dy}{d\xi} + 0.6\xi}{3y}.$$

The boundary conditions for the ODE are as follows:

$$\frac{dy}{d\xi}\bigg|_{\xi=0} = -0.75 \frac{q_0}{\sqrt[3]{y(0)}} \quad \text{at the inlet } \xi = 0 \quad (15)$$

$$y(\xi_*) = 0 \text{ at fluid front } \xi = \xi_* \quad (16)$$

$$\text{and with a speed equation of } \frac{dy}{d\xi}\bigg|_{\xi=\xi_*} = -0.6\xi_* \text{ at the fluid front } \xi = \xi_* \quad (17)$$

With the addition of the speed equation, there are two, rather than one, boundary conditions at the fluid front for the governing second-order ODE. From a mathematical point of view, this boundary value problem (BVP) is ill-posed in the Hadamard sense (Hadamard, 1902; Tikhonov, 1963). By solving the self-similar BVP problem using finite differences without regularization, the implementation failed to obtain more than two correct digits ($\xi_* = 1.01$, $\psi(0) = 0.83$). Moreover, the results strongly deteriorated near the crack tip when a fine mesh was used. Thus, proper regularization is required for more accurate and stable numerical results. Linkov suggested a regularization technique by imposing a new boundary condition at a small distance ε behind the front instead of on the front itself. The new boundary condition was formulated by the combination of the speed equation (17), and the

boundary condition (15) and is expressed as

$$y(\xi_e) = 0.6\xi_e^2(1 - \varepsilon) \quad (18)$$

The BVP expressed in (14), (16), and (18) is well-posed and can be solved by finite differences. Unlike the solution without regularization, results have been obtained with seven correct significant digits ($\xi_* = 1.0073486$, $\psi(0) = 0.8390285$).

Mishuris et al. (2012) investigated the efficiency of the method proposed by Linkov (2011c) for the PKN problem for cases with and without leak-off on the aspects of the stiffness properties, computational accuracy, and solution sensitivity. The study showed that the stiffness of the ODE obtained after spatial discretization significantly decreased by assuming a fixed particle velocity at an iteration. Moreover, utilizing the speed equation rather than a common global mass balance increased the accuracy of time-stepping Crank–Nicolson schemes.

Kusmierczyk et al. (2012) further improved the modified formulation and built a general numerical framework for different leak-off regimes (vanishing regime, bounded regime, and crack tip singularity). The enhanced reformulation was based on the employ of a new dependent variable, which was an integral of the crack opening, expressed as

$$\omega(t, x) = \int_x^1 w(t, \xi) d\xi \quad (19)$$

This dependent variable $\omega(t, x)$ indicates the fracture volume estimated from the fracture tip and is not dependent on any particular type of asymptotic behavior in the tip region; the form of the governing equations of ω is the same regardless of $w(x, t)$ asymptotics. An analysis based on the reformulated system in this study demonstrated that the stiffness properties and the computational accuracy can be considerably affected by the selection of dependent and independent variables. The use of ω resulted in a slightly higher stiffness of the system (i.e., one order of magnitude higher) and considerable accuracy of computations.

Wrobel and Mishuris (2013) presented a novel and efficient algorithm for the PKN model based on the pseudo-spectral approach. The algorithm was employed for a self-similar problem and extended to a general time-dependent formulation. The fundamental difference between these two problems was that in the latter, the fracture length $l(t)$ became an element of the solution. No regularization technique was required for this method, as the exact boundary conditions at the crack tip were imposed directly into the integration schemes. The respectively developed solvers for both cases (self-similar and time-dependent) provided better computational accuracy at a much lower computation cost (in terms of the reduction of densities of the temporal and spatial meshes) than the previous method of Mishuris et al. (2012).

Linkov (2013, 2014) focused on non-Newtonian fluid by employing a modified formulation (2011c) for a shear-thinning fluid ($0 < n < 1$, where n is the flow behavior index), commonly used in hydraulic fracturing. The derived analytical solutions in self-similar variables for perfectly plastic ($n = 0$) and Newtonian ($n = 1$) fluids implied that the self-similar solution for shear-thinning fluids was practically independent of the fluid behavior index. This led to simple universal formulas for the self-similar fracture length, fracture width, particle velocity, fluid pressure, and fluid flux. He also introduced the concept of fluid equivalence in terms of the fracture length at a given time and pumping rate. The analysis showed that, for the equivalent shear-thinning fluids (i.e., fluids producing hydraulic fractures of the same fracture length for a given time), the differences in the physical quantities (i.e., fracture length, speed, opening, and net pressure) were inconsiderable within a pumping time of between 10s and 27h. As a result, it was recommended that when fluids are equivalent according to the suggested criterion, a fracturing fluid should be selected based on other aspects such as technology, economy, and environment.

Wrobel and Mishuris (2015) revisited the particle velocity approach

for various hydraulic fracturing models, including the PKN model. They introduced a new dependent variable called the reduced particle velocity and reformulated the governing equations in terms of this dependable variable and fracture opening. The reduced particle velocity is expressed as

$$\varphi(t, x) = \nu(t, x) - x\nu_0(t) \quad (20)$$

where ν_0 is the first term in the asymptotic solution of the particle velocity ν , which is expressed as

$$\nu(t, x) = \nu_0(t) + \nu_1(t)(1 - x) + o((1 - x)^2), \quad x \rightarrow 1 \quad (21)$$

This numerical scheme was implemented for both reduced self-similar and general time-dependent formulations of the problem. The resulting accuracy in terms of relative errors of the crack opening $\delta\hat{w}$ and the reduced particle velocity $\delta\hat{\varphi}$ in the self-similar solutions and the crack length δL in the time-independent solution was analyzed, as shown in Fig. 3. Here the results for the studied case of non-zero leak-off (benchmark I) are compared with the no leak-off solution (benchmark II) given by Mishuris et al. (2012) and both leak-off and no-leak-off efficient solvers (dashed line) given by Wrobel and Mishuris (2013). The latter did not employ the reduced particle velocity and required additional post-processing; thus, $\delta\hat{\varphi}$ was not calculated. The relative errors were calculated for a range of 10 to 300 nodal points, considering the density of spatial mesh was increased at the end of the interval. The resulting graphs indicate that the resulting error depended on the benchmark type. Employing benchmark II reduced the error caused by the numerical integration of the leak-off. The solution accuracy improved with increasing mesh density, although the error reduction decreased with an increased number of nodal points, N . The comparison with the results that used efficient integral solvers showed that the latter provided better accuracy for small and large N and large M . However, the new algorithm based on the reduced particle velocity did provide higher accuracy in the case of leak-off at an intermediate interval ($N \approx 50 - 150$). The resulting error of the evolution of fracture length indicates that the accuracy of the crack length mainly depended on the density of spatial meshing and was insensitive to the time step density.

Perkowska et al. (2016) extended the proposed universal particle velocity-based algorithm (Wrobel and Mishuris, 2015) to include shear-thinning fluids ($0 < n < 1$). Their work demonstrated that the average computational errors of the crack opening, particle velocity, and reduced particle velocity were relatively insensitive to the value of the fluid behavior index. For a fixed n , the solution accuracy in terms of the average crack opening, particle velocity, and reduced particle velocity increased proportionally with the number of nodal points N . The errors were no longer reduced when $N > 150$ for fluids with $n \geq 0.5$. This computational scheme, thus, was shown to provide more accurate numerical results than any other data available in the literature. In particular, when $N = 100$, at least three-order higher accuracy was obtained regardless of the considered hydraulic fracture model.

3.4. Group 4: models accounting for the turbulent effect

The models discussed so far in groups 1–3 have assumed a laminar flow of the hydraulic fluid. Hence, this section summarizes PKN models that consider the effect of turbulence for the application of low-viscosity fluid with a high injection rate recently used for the well stimulation technique of the petroleum industry.

The pioneering work of Perkins and Kern (1961) did include a solution of the fracture width for a turbulent flow regime. They chose a Fanning friction factor of 0.0125 (i.e., Reynolds number ≥ 2500) for a fracture corresponding to a relative roughness of the walls of 0.02. This simple model assumed no leak-off and only derived the crack width equation for the turbulent flow regime of Newtonian fluids (together with laminar flow of Newtonian and non-Newtonian fluids). The effect of a turbulent flow regime in hydraulic fracturing modeling was not

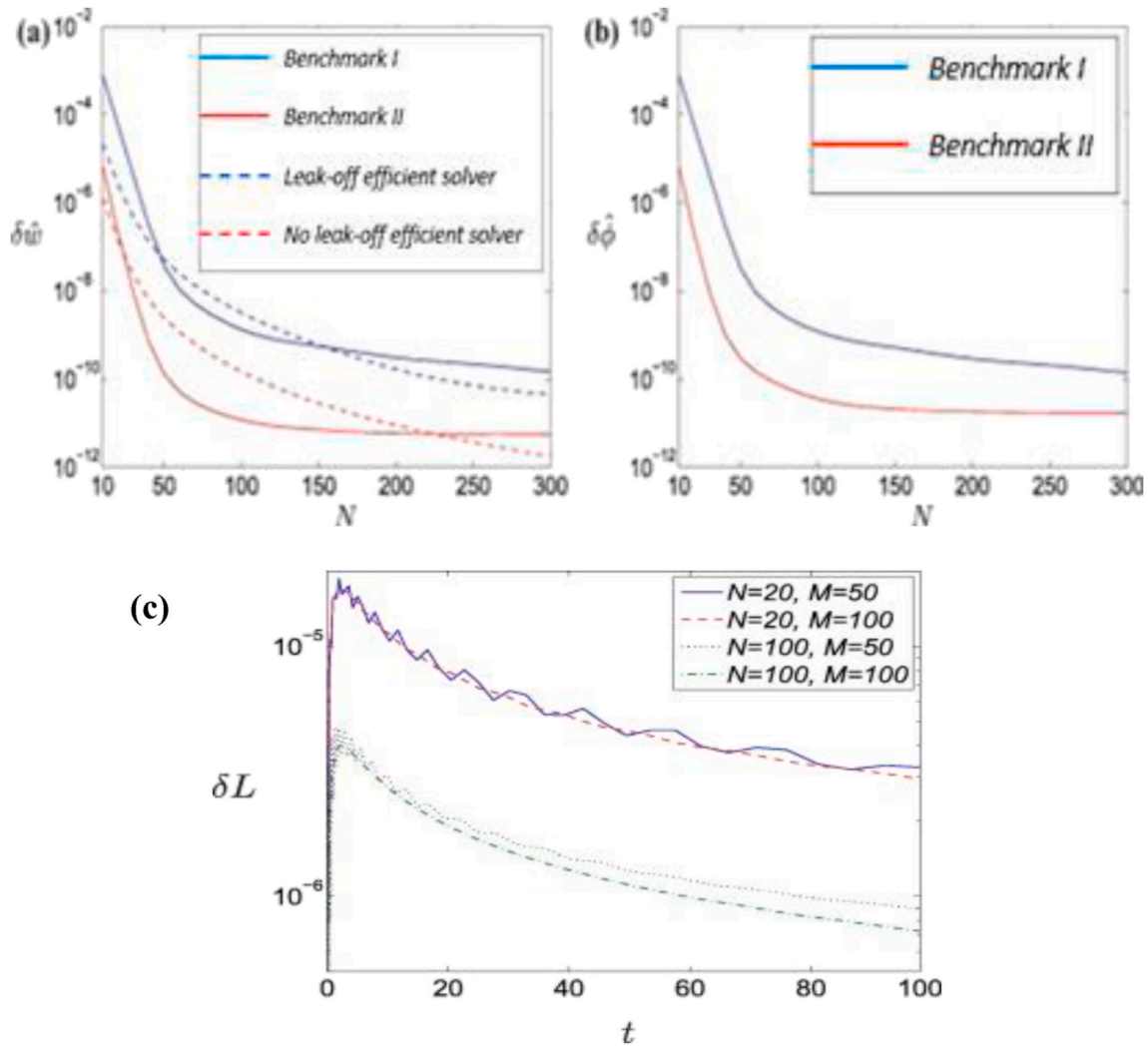


Fig. 3. The relative errors of the self-similar solution for (a) the crack opening $\delta \hat{w}$, (b) the reduced particle velocity $\delta \hat{\phi}$, and (c) the crack length error δL for various spacial and temporal meshing (where benchmark I represents the case of non-zero leak-off, benchmark II represents the case of no leak-off, N denotes the number of nodal points, and M denotes the number of time steps). Dashed lines in (a) correspond to the results obtained by integral solver described in Wrobel and Mishuris (2013). (Modified from Wrobel and Mishuris (2015)).

revisited until Anthonyrajah et al. (2013) considered a turbulent flow for blade-shaped (PKN) hydraulic fractures in the case of zero leak-off in 2013. Here, the mathematical governing equations were derived based on the work of Emerman et al. (1986); analytical solutions for two special and non-physical cases involving constant propagation speed and constant total fracture volume were presented, and a numerical solution method for general injection was demonstrated. Later, Newman (2016) extended this model of the PKN problem using Prandtl's mixing length. Ames and Bunger (2015) evaluated the role of a turbulent fluid flow when low-velocity slickwater was used in hydraulic fracturing for unconventional reservoirs. They derived scaling laws for the propagation of a PKN-type fracture in a fully turbulent regime. The turbulent fluid flow equation was based on the work of Tsai and Rice (2010):

$$q_x = \left(\frac{\delta p_{net} 4w^3}{\delta x \rho f_D} \right)^{\frac{1}{2}} \quad (22)$$

where ρ is fluid density and f_D is the Darcy friction factor, which was approximated by the Gaukler–Manning–Strickler (GMS) (Manning, 1891; Strickler, 1981) equation as

$$f_D = 0.143 \left(\frac{k}{w} \right)^{\frac{1}{3}} \quad (23)$$

where k represents the wall roughness.

This resulted in a fracture with a shorter length and greater width and pressure when compared with the laminar flow solution. Kano et al. (2015) developed an approximate analytical solution for a fully turbulent rough-walled hydraulic fracture with high leak-off using the GMS approach. Zolfaghari et al. (2016) also presented a semi-analytical solution for the PKN model considering turbulent flow without leak-off using the GMS friction factor. Their results indicated a substantial overestimation of the fracture length and an underestimation of the fracture opening and pressure by 40%–50% were incurred when a laminar flow regime, rather than a turbulent flow regime, was used under conditions where $Re > 2500$.

Although these researchers (Anthonyrajah et al., 2013; Ames and Bunger, 2015; Kano et al., 2015; Zolfaghari et al., 2016) employed a fully turbulent, rough flow regime over the entire fracture by utilizing the GMS friction factor, Dontsov (2016) reported that this assumption was not always valid. As shown by Nilson (1981), the Reynolds number is proportional to the crack opening, which disappears at the crack tip. Therefore, the flow is always laminar in the near-tip region. To capture the turbulent, laminar, and transition regimes, Dontsov adapted the empirical Churchill formula (1977) for the friction factor of pipes to the friction for fractures (i.e., flow between two parallel plates). This

Table 1

Summary of the progress in modeling and computational methods for solving PKN-type problems.

Group	Progress in modeling and computational methods	References
Group 1: Classical model	Formulating fracture width for PKN model Adding Carter's leak-off and storage Adding Stefan condition	Perkins and Kern (1961) Nordgren (1972) Kemp (1990)
Group 2: Fluid flux approach	Evaluating poroelastic effect using an influence function Evaluating poroelastic effect using pressure-dependent leak-off Proposing a non-local elasticity equation Reexamining PKN model using scaling arguments and crack tip asymptotics Approximating the crack opening and fluid flux by a cubic polynomial over a discretization interval Employing three different toughness criteria	Detournay et al. (1990) Abousleiman (1991) Adachi and Peirce (2008) Kovalyshen and Detournay (2009) Adachi et al. (2010)
Group 3: Particle velocity approach	Utilizing particle velocity and a regularization technique Computational efficiency in terms of the stiffness, accuracy, and sensitivity of Linkov's approach Proposing a dependent variable as an integral of the crack opening Presenting an efficient algorithm based on the pseudo-spectral approach Employing Linkov's method for shear-thinning fluids Employing a dependent variable of reduced particle velocity (for Newtonian fluids)	Dontsov and Peirce (2016) Linkov (2011a, 2011b, 2011c, 2012a, 2012b) Mishuris et al. (2012) Kusmierczyk et al. (2012) Wrobel and Mishuris (2013) Linkov (2013, 2014) Wrobel and Mishuris (2015)
Group 4: Turbulent effect	Extending Wrobel and Mishuris's method for shear-thinning fluids Fracture width solution using a Fanning friction factor of 0.0125 Zero leak-off solution based on the work of Emerman Investigating the effect of a turbulent regime using the GMS (Gaukler–Manning–Strickler) friction factor Adapting the Churchill formula for the friction factor for turbulent, laminar, and transition regimes Employing Blasius and GMS friction factors for fully smooth and fully rough turbulent regimes	Perkowska et al. (2016) Perkins and Kern (1961) Anthonyrajah et al. (2013); Newman (2016) Ames and Bunker (2015); Kano et al. (2015); Zolfaghari et al. (2016) Dontsov (2016) Zia and Lecampion (2016, 2017)

formula is expressed as

$$f_D = 8 \left(\left(\frac{12}{Re} \right)^{12} + (A + B)^{-1.5} \right)^{\frac{1}{2}} \quad (24)$$

where

$$A = \left(2.457 f_0 \left(Re, \frac{k}{D} \right) \right)^{16}, \quad B = \left(\frac{37530}{Re} \right)^{16}, \quad \text{and} \quad (25)$$

$$f_0(Re, r) = \left| \log \left(\left(\frac{7}{Re} \right)^{0.9} + 0.27r \right) \right|, \quad r = \frac{k}{2w} \quad (26)$$

where $Re < 2000$ represents the laminar regime, $Re > 3000$ represents the turbulent regime, $2000 < Re < 3000$ represents the transition regime, and D is the pipe diameter.

In this work, Dontsov also derived an asymptotic solution for the fully turbulent regime and studied the influence of the transition regime on the tip region.

Zia and Lecampion (2016, 2017) employed the concept of equivalent laminar proposed by Jones (1976) to investigate the influence of turbulence on the propagation of the PKN model. An equivalent Reynolds number $\overline{Re}$ for the PKN model was defined as

$$\overline{Re} = \frac{5}{\pi} Re \quad (27)$$

where $Re = \frac{\rho \bar{w} \bar{v}}{\mu}$ is the Reynolds number computed using the averaged fracture width $\bar{w}$ and $\bar{v}$ is the average fluid velocity.

Using this equivalent Reynolds number, the laminar friction factor $f_L = 16/\overline{Re}$. For fully smooth and fully rough turbulent regimes, the authors used Blasius (1913) and the GMS friction factors, respectively:

$$f_B = f'_B \overline{Re}^{-\frac{1}{4}}, \quad f'_B = 0.316 / 4 \quad (28)$$

$$f_M = f'_M \left(\frac{k}{w} \right)^{\frac{1}{3}}, \quad f'_M = 0.143 / 4 \quad (29)$$

The laminar to smooth or rough turbulent transition regimes were mapped using the method proposed by Yang and Dou (2010).

Zia and Lecampion numerically solved the transition regime from laminar to turbulent flow as a function of Reynolds number at the hydraulic fracture. They showed that in practice, the Reynolds number can be less than 5000, which is within the laminar to turbulent transition regime. They also derived limiting solutions for turbulent smooth and rough cases and demonstrated that the Reynolds number at the inlet must be at least 10000 to apply a fully turbulent rough solution. The effect of additional friction-reducing agents (which are commonly employed in high-rate hydraulic fracturing) was also evaluated (Zia and Lecampion, 2017). Their results showed that the fracture length and inlet width were only approximately 10% – 15% different from the fully laminar solution (when $Re = 10^4$). In the case of no additional friction reducers, the difference was approximately 40%.

A summary of the discussed contributions to the PKN problem is shown in Table 1. The main findings are listed to show the research flow and how the solutions have been developed over time.

4. Computational challenges

Despite the assumptions made to simplify the PKN model, solving the fracture propagation of the PKN problem is still a challenging task. The main computational challenges are owing to the strongly nonlinear lubrication PDE, moving boundary characteristics, and degenerate coupled governing equations at the crack tip.

Analytical solutions at the crack tip in the literature (Desroches et al., 1994) has proved that, under the assumption that fluid reaches the tip (no lag), the strong non-linearity of the coupling governing system leads to a singularity of fluid pressure and fracture opening at the tip region. This singularity is weaker than the classical inverse square root (i.e., $w \propto \frac{K_{IE}^{\frac{1}{2}}}{E^{\frac{1}{2}}}$) predicted by linear elastic fracture mechanics. Consequently, the crack tip velocity, which is proportional to the product of the fracture width and pressure gradient (i.e., $v \propto 0^* \infty$), cannot be accurately estimated (Peirce and Detournay, 2008). As a result, it is difficult to determine the crack tip position by the crack tip velocity.

One proposed way to overcome this is to explicitly employ the particle velocity and the speed equation (Linkov, 2011c). A new boundary condition is created by combining the speed equation and one boundary

condition at the inlet. The governing equations are also reformulated using the particle velocity and the crack opening taken to a degree as primary physical quantities. The modified formulation has been demonstrated as well-posed and avoiding singularity terms at the crack tip region. This method may also reduce the stiffness of the governing ODE system after spatial discretization.

In the tip region, the Reynold lubrication equation, represented as a combination of the Poiseuille equation and the continuity equation, might become problematic when solved numerically, due to the degeneracy of the lubrication equation at the crack tip and the difference in length scales between this region and the overall length of the fracture. Both explicit and implicit numerical schemes have been successfully used for solving the problem.

Essentially, explicit methods require that the chosen time step is not greater than a critical value for satisfying the stability condition; because only results from the previous step are used in the following step, the total numerical error can be too large if the time step is too large. Therefore, in the case of explicit schemes, any differencing technique chosen requires time steps small enough to meet the highly restrictive Courant–Friedrichs–Lewy stability condition while ensuring reasonable computational time. Linkov (2019) has demonstrated that this task is achievable. An implicit scheme can use a large time step; however, it requires proper preconditioners to provide convergence within an accepted number of iterations. An implicit finite difference technique with no restriction in the time step can be employed to overcome this challenge, as was shown in the work of Adachi et al. (2010).

5. Conclusion

This review summarized the evolution and improvement of the PKN-type model used for modeling hydraulic fracturing. The considered PKN models were divided into four groups including the classical models, those employing the fluid flux approach, those following the particle velocity, and those accounting for the turbulent effect of the fluid flow. The main findings of each model were discussed to provide readers with an overview of the PKN problem. The interaction among fundamental processes in hydraulic fracturing can make modeling challenging, even for the seemingly simple PKN geometry. Mathematical formulations of the problem based on the fluid flux or the particle velocity have shown their advantages and limits.

As mentioned in section 1, PKN type models serve as a base for more complex models. The PKN model itself is not meant for practical application owing to its modeling simplifications. An efficient-computational numerical solution for PKN model that accounts for a non-local elasticity equation rather than the local elasticity equation should be considered to provide a complete set of benchmarks and verification. Moreover, the effect of hydraulic fracture clustering from one perforation point, as observed in the HFTS field-based project, can be analyzed and included in the modeling.

Declaration of competing interest

There are no conflicts of interest to declare.

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